

SMILE SHARMA

Software Engineer | Distributed Systems & Generative AI

Noida, India • +91-8741865218 • smilesbelley270702@gmail.com

linkedin.com/in/smilesharma27 • github.com/Smilegits • Portfolio • Aperture

Professional Summary

Software Engineer with 2+ years of experience designing and shipping production-grade distributed systems and Generative AI applications. Proficient in Java, Spring Boot, and Python across microservices, event-driven pipelines (Apache Kafka, RabbitMQ), and cloud platforms (AWS, Azure, GCP). Proven record of raising pipeline reliability to 99.9% and reducing execution time by 40–50% through CI/CD automation. Experienced in LLM/RAG systems, Kubernetes, and full lifecycle ownership from design to deployment.

Technical Skills

Languages: Java, Python, SQL, JavaScript (ES6+), C++, C#

Backend & Frameworks: Spring Boot, Spring Framework, RESTful APIs, Microservices, Node.js, React.js

Distributed Systems & Messaging: Apache Kafka, RabbitMQ, Distributed Systems Design, System Design, Cassandra, Redis

Databases: PostgreSQL (PPAS), AlloyDB, MongoDB, MySQL, SQL

Cloud: AWS (EC2, S3, Lambda), Microsoft Azure (Azure OpenAI, Azure ML), Google Cloud Platform (Vertex AI)

DevOps & CI/CD: Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD Pipelines, Blue-Green Deployments, LaunchDarkly, Git, Agile/Scrum

AI/ML & Generative AI: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, PyTorch, TensorFlow, Transformers, Model Inference

Observability & Security: OpenTelemetry, Prometheus, Grafana, OAuth, Credential Hardening

Core CS: Data Structures, Algorithms, Object-Oriented Programming (OOP), System Design

Professional Experience

UKG (Ultimate Kronos Group) — Software Engineer

Aug 2024 – Present

Noida, India | Healthcare Analytics (HCA), Distributed Systems, Cloud & DevOps

- Designed and deployed **distributed data pipelines** using **RabbitMQ and Cassandra** to process high-volume payroll and healthcare metrics, sustaining **99.9% reliability** at enterprise scale.
- Developed **Java and Spring Boot microservices** with **RESTful APIs** integrated into enterprise payroll platforms, managing contract-employee data for large hospital systems.
- Architected automated workflows and **CI/CD pipelines** that reduced execution time from 2–3 days to 10 hours — a **40–50% operational efficiency gain**.
- Implemented **Data Lifecycle Cloning (DLC)** on the HCA service to resolve data-refresh issues for large customers (Trinity Health, Banner), coding table-clone filtering logic to exclude audit logs and staging data during tenant cloning.
- Optimized tenant provisioning with a **LaunchDarkly feature flag** for single-DB-node async provisioning; built, patched, and validated the feature end to end.
- Executed **Blue-Green deployments** and orchestration migrations for the Pro WFM Healthcare Productivity Service using **Kubernetes and Platform Workflow Orchestrator (PWO)**, verifying releases via Express Upgrade workflows.
- Hardened security under Project Sphere by eliminating hardcoded **RabbitMQ credentials** across HCA and WFM services (UKG Security Standards compliance); delivered **OAuth** token generation and rebranding verification for HCA tenants.
- Built scalable reporting modules in Python and SQL analyzing patient-volume metrics across **10+ hospitals** and **20+ automated reports** to drive data-driven healthcare decisions.
- Engineered a high-performance **data-purger framework** in Spring Boot that archived **5M+ records (~2 TB)** and cut database storage footprint by **~35%**, improving query performance by **~40%** while ensuring regulatory compliance.

GKMT IT — Frontend Developer (Internship)

2023

- Built **10+ responsive React.js interfaces** from design specs during a 6-month internship, cutting page load time by **~25%** via component reuse and lazy loading, and reducing UI defects by **~20%** through code reviews and standardized Git workflows.

Projects

Aperture — Token-Efficient Document Intelligence Platform

Live

Java, Spring Boot, Python, Apache Kafka, Redis, PostgreSQL, MongoDB, Kubernetes, RAG, LLMs

- Architected a **multi-tenant RAG platform** across 6 polyglot microservices with **Kafka-driven asynchronous ingestion** — idempotent consumers, dead-letter queues, and replay-based reindexing.

- Reduced **LLM token consumption per query** via a version-tagged **Redis semantic cache**, cross-encoder reranking with context compression, and a confidence-gated model cascade.
- Built a per-tenant token/cost ledger in PostgreSQL with budget enforcement and a CI-integrated evaluation harness scoring faithfulness and context precision on a 150-question golden set.
- Instrumented end-to-end **OpenTelemetry** tracing with **Prometheus/Grafana** dashboards for cost, cache hit rate, and p95 latency.

Generative AI Chatbot — Google Gemini Pro

Python, Google Gemini Pro, Streamlit, PyTorch, LLMs, Prompt Engineering

- Built a context-aware conversational AI on **Gemini Pro** with optimized **prompt engineering** for low-latency responses; implemented conversation-history and state management over Transformer architectures.

Dynamic Blog Application — Full-Stack

React.js, JavaScript, RESTful APIs, Responsive Design

- Developed a full-stack CRUD blog with **React Router** client-side routing and custom hooks consuming REST APIs, following component-based architecture and responsive UI/UX.

Education

Dr. B. R. Ambedkar National Institute of Technology (NIT Jalandhar)

2020 – 2024

B.Tech, Information Technology • CGPA: 8.09 • Jalandhar, India

Certifications

- **Build Real-World AI Applications with Gemini & Imagen** — Google Cloud
- **Prompt Design in Vertex AI** — Google Cloud
- **AWS Educate: Machine Learning Foundations** — Amazon Web Services
- **AWS Cloud Quest: Cloud Practitioner** — Amazon Web Services
- **AWS Knowledge: Cloud Essentials** — Amazon Web Services